

Project Closure Report

Advanced Chooser Option Pricing Model

Underlying JPMorgan Chase and Co
Project period Week 1 through Week 8
Closure date 2026-09-16

Closure Summary

The eight-week project produced an end-to-end chooser-option research pipeline that connects market data collection, preprocessing, Black-Scholes-Merton pricing, model validation, machine learning, sensitivity analysis, and an interactive Streamlit prototype. The technical work is complete and archived. The narrated demonstration has been recorded for separate delivery to the supervisor. The brief requires a deployable tool and GitHub README, not public cloud hosting.

The primary route forecasts volatility with Random Forest and applies the chooser BSM formula, reducing historical test proxy RMSE by 11.30% versus BSM. After a live negative-price failure, bounded time-value regression replaced the direct linear model. Its revised test RMSE is 4.5404 and MAE 2.8501; Approach 1 remains primary by RMSE.

Project Objectives and Outcomes

No	Objective	Closure outcome
1	Replicate and validate a BSM chooser model	Completed with paper-matched parameters and numerical checks
2	Build a multidimensional market dataset	Completed using JPM, VIX, and Treasury data from 2018 to 2024
3	Develop machine-learning extensions	Completed through volatility forecasting and direct pricing routes
4	Compare ML with the BSM baseline	Completed on chronological test data; direct revision is retrospective
5	Package a usable pricing tool	Complete; runnable tool and README published in the GitHub repository

Eight Week Delivery Record

Period	Archived work	Status
Week 1	Data specification, API notebook, raw JPM, VIX, and Treasury data	Complete
Week 2	Cleaned 23-column dataset, automated preprocessing, engineered features	Complete; scheduled GitHub Actions passed; news-sentiment gap recorded
Week 3	Chooser BSM notebook, validation notebook, parameter configuration	Complete
Week 4	Baseline evaluation, validation report, performance benchmark, reusable BSM code	Complete with transparent proxy and listed-option validation
Week 5	ML architecture, optimized features, two initial modeling approaches	Complete
Week 6	Trained .pkl and .joblib models, performance comparison, SHAP analysis	Complete
Week 7	Sensitivity report, stress analysis, pricing prototype, complete feature updater	Complete with documented sentiment-data limitation
Week 8	Packaged pricing tool, final technical report, presentation deck	Tool, reports, presentation and recording complete; video delivered separately

Data and Model Chain

- Week 1 raw market data flows into the Week 2 preprocessing and feature pipeline.
- Week 3 supplies the validated chooser BSM implementation and contract configuration.
- Week 4 evaluates that model historically and under listed-option sanity checks.
- Weeks 5 and 6 train, select, compare, and serialize the ML extensions.
- Weeks 7 and 8 integrate the revised direct model, regenerated sensitivity analysis, complete feature refresh, presentation and interface. Week 1–5 inputs and Approach 1 predictions are unchanged. The historical test is reused retrospectively for this repair. Direct validation RMSE worsened from 5.4837 to 7.3983, so improvement is not uniform.

Key Technical Results

Evidence	Result	Interpretation
Processed market dataset	1,760 rows and 23 columns	Aligned JPM, VIX, and Treasury history
Paper choice decisions	10 of 10 matched	Week 3 replication check
Put-call parity error	0.000000	Analytical consistency check
Monte Carlo comparison	0.823 standard errors	Closed-form price within simulation uncertainty
Selected ML route	RMSE 4.3203 and R2 0.9575	Random Forest volatility plus BSM
BSM baseline	RMSE 4.8706 and R2 0.9460	Same 252 historical test dates
Direct-price model	RMSE 4.5404 and R2 0.9530	Repaired with structural bounds

Model Delivery and Reproducibility

Week 6 now includes native Python pickle files for both selected estimators, as required by the project outline. Each file was reloaded and checked on all 252 historical test rows. Joblib copies are retained for the packaged Week 8 application, and model metadata records the feature order, contract assumptions, evaluation dates, artifact manifest, and round-trip validation results.

Stress Test Record

Scenario	Chooser price	Change
Base	\$99.2617	\$0.0000
50 percent volatility spike	\$115.6683	+\$16.4066
Two percentage point rate hike	\$101.3317	+\$2.0700
Combined stress	\$116.9177	+\$17.6561

Limitations and Closure Controls

Limitation	Control
No public historical chooser transaction series	All chooser accuracy scores are labeled as theoretical proxy results
CME wording does not align cleanly with a JPM equity chooser	Cboe listed JPM vanilla quotes are used only as a BSM market sanity check
No reproducible news sentiment series	The gap is disclosed; VIX is not relabeled as sentiment
Dividend yield was not collected historically	The paper value of 2.33 percent remains an explicit assumption
Tree models have weak extrapolation	BSM remains visible and the interface warns outside fitted ranges
Serialized estimators require a trusted environment	Load only project-owned files and preserve documented package versions

Archive Location

All formal reports are consolidated under the project Reports directory, organized by weekly subfolders. This closure report is stored at the Reports root for direct archival access. Source code, datasets, notebooks, model artifacts, figures, and the deployable application remain in their week-specific project directories.

GitHub Delivery

The project repository is <https://github.com/Frank-Wang-Jc/jpm-chooser-option-pricing>. The latest inspected project-validation, weekly-preprocessing and market-refresh runs all passed on GitHub Actions, verified on 16 September 2026. The runnable tool and README meet the brief's deployable-tool requirement.

Outstanding Submission Actions

- Send the recorded demonstration directly to the supervisor, separately from the repository.
- Submit the final report PDF and presentation deck with the repository link.
- Before submission, verify reviewer access to the GitHub repository. Credentials, local environments, licensed references, and duplicate archives remain excluded.

Closure Statement

The project has reached technical completion across the planned eight-week sequence. Its code, data, validation evidence, trained models, reports, presentation, and local application are organized for review and archival. The demonstration is recorded for separate delivery; confirm repository access and submit the final package. Public application hosting is optional.